Applications in Precision Medicine

- Identify new disease genes
- Genomics-based diagnostics
- Precision treatment

Structural variation and human diseases

1. Identify new disease genes

Trait	Gene/locus	Reference
Psoriasis	*beta-defensin*	Hollox et al, 2008
Emphysema	5p15.33, 7q22.1	Choi et al, 2009
Acute myocardial infarction and stroke (smokers)	*C4B* encoding C4	Blasko et al., 2008
Alveolar capillary dysplasia	*FOXF1*	Stankiewicz et al, 2009
Hypertrichosis	17q24.2-q24.3	Sun et al, 2009
Chronic pancreatitis	*PRSS1*	Le Marechal et al, 2008
Lupus with glomerulonephritis	*FCGR3B*	Aitman et al, 2006
Crohn disease	*HBD-2 (beta-defensin)*, *IRGM*	Fellermann et al, 2006 McCarroll et al, 2008
Heterozygous familial hypercholesterolemia	*LDLR*	Pollex et al, 2007
Atherosclerosis	*LPA*	Pollex et al, 2007
Hemophilia A	*Factor VIII*	Lakich et al, 1993
Malaria susceptibility	*a-globin*	Flint et al, 1986
Systemic lupus erythematosus	*Complement C4 component, FCGR3B*	Fanciulli et al, 2007 Willcocks et al, 2008
HIV susceptibility	*CCL3L1*	Gonzalez et al, 2005

2. Genomics -based screening

重新查询 **北京市新生儿耳聋基因筛查报告单** 下载报告 打印报告

检测结果

序号	基因名称	突变位点	检测结果	结果说明	检测结果图
1	GJB2	35 del G	野生型	正常	
2	GJB2	176 del 16	野生型	正常	
3	GJB2	235 del C	野生型	正常	
4	GJB2	299 del AT	杂合突变型	异常	
5	GJB3	538 C > T	野生型	正常	
6	SLC26A4	2168 A > G	野生型	正常	
7	SLC26A4	IVS7-2 A > G	野生型	正常	
8	线粒体12s rRNA	1494 C > T	野生型	正常	
9	线粒体12s rRNA	1555 A > G	野生型	正常	

筛查结果 未通过

【建议】

未通过本次相关耳聋基因筛查，此结果异常并非最终诊断，需要进一步的听力学随访和耳聋基因诊断，请于工作时间致电下方医院，预约就诊时间。

【提示】

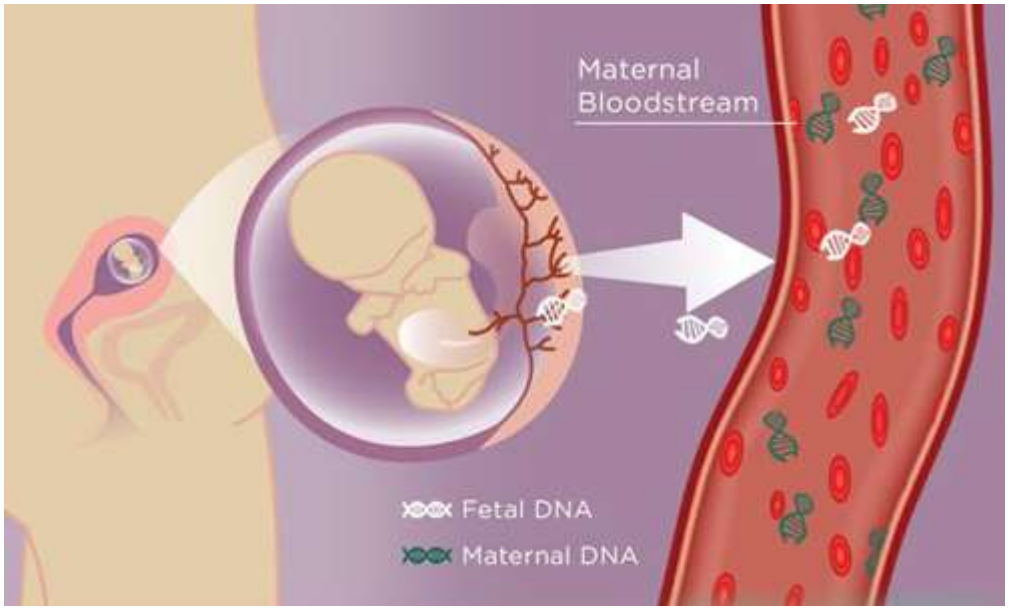
此结果仅对本份检测标本负责！由于儿童耳聋发病原因的复杂性，此结果仅是对中国常见的4个耳聋相关基因（GJB2、GJB3、SLC26A4及线粒体12SrRNA）的9个突变位点进行筛查，因此，即便是筛查结果为通过，如在发现孩子听力异常时也应及时就医。

【遗传咨询医院】

医院名称	医院地址	预约就诊电话	就诊须知
首都医科大学附属北京同仁医院	北京市东城区东交民巷1号（崇文门）	58265831	请持听力筛查和基因筛查报告单，按预约时间到北京同仁医院西院4层耳科门诊挂号就诊。

Non-Invasive Prenatal Testing (NIPT)

2. Genomics -based screening



3. Precision treatment

The NEW ENGLAND JOURNAL of MEDICINE

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70-Gene Signature as an Aid to Treatment Decisions in Early-Stage Breast Cancer

F. Cardoso, L.J. van't Veer, J. Bogaerts, L. Slaets, G. Viale, S. Delaloge, J.-Y. Pierga, E. Brain, S. Causeret, M. DeLorenzi, A.M. Glas, V. Golfinopoulos, T. Goulioti, S. Knox, E. Matos, B. Meulemans, P.A. Neijenhuis, U. Nitz, R. Passalacqua, P. Ravdin, I.T. Rubio, M. Saghatchian, T.J. Smilde, C. Sotiriou, L. Stork, C. Straehle, G. Thomas, A.M. Thompson, J.M. van der Hoeven, P. Vuylsteke, R. Bernards, K. Tryfonidis, E. Rutgers, and M. Piccart, for the MINDACT Investigators*

Early Breast Cancer Treatment

Breast cancer staging

Stage	Primary Tumor	Nodes	Metastases
Stage 1A	≤ 20 mm	None	None
Stage 1B	≤ 20 mm	Nodal Micrometastases (>0.2 mm <2.0 mm)	None
Stage IIA	≤ 20 mm > 20 mm ≤ 50 mm	N1 None	None None
Stage IIB	> 20 mm ≤ 50 mm > 50 mm	N1 None	None
Stage IIIA	≤ 50 mm > 50 mm	N2 N1 or N2	None
Stage IIIB	Extension to chest wall and/or skin	N0 - N2	None
Stage IIIC	Any size	N3	None
Stage IV	Any size	Any involvement	Detectable

Treatment for early breast cancer usually involves some combination of surgery, radiation therapy, chemotherapy, hormone therapy and/or targeted therapy.

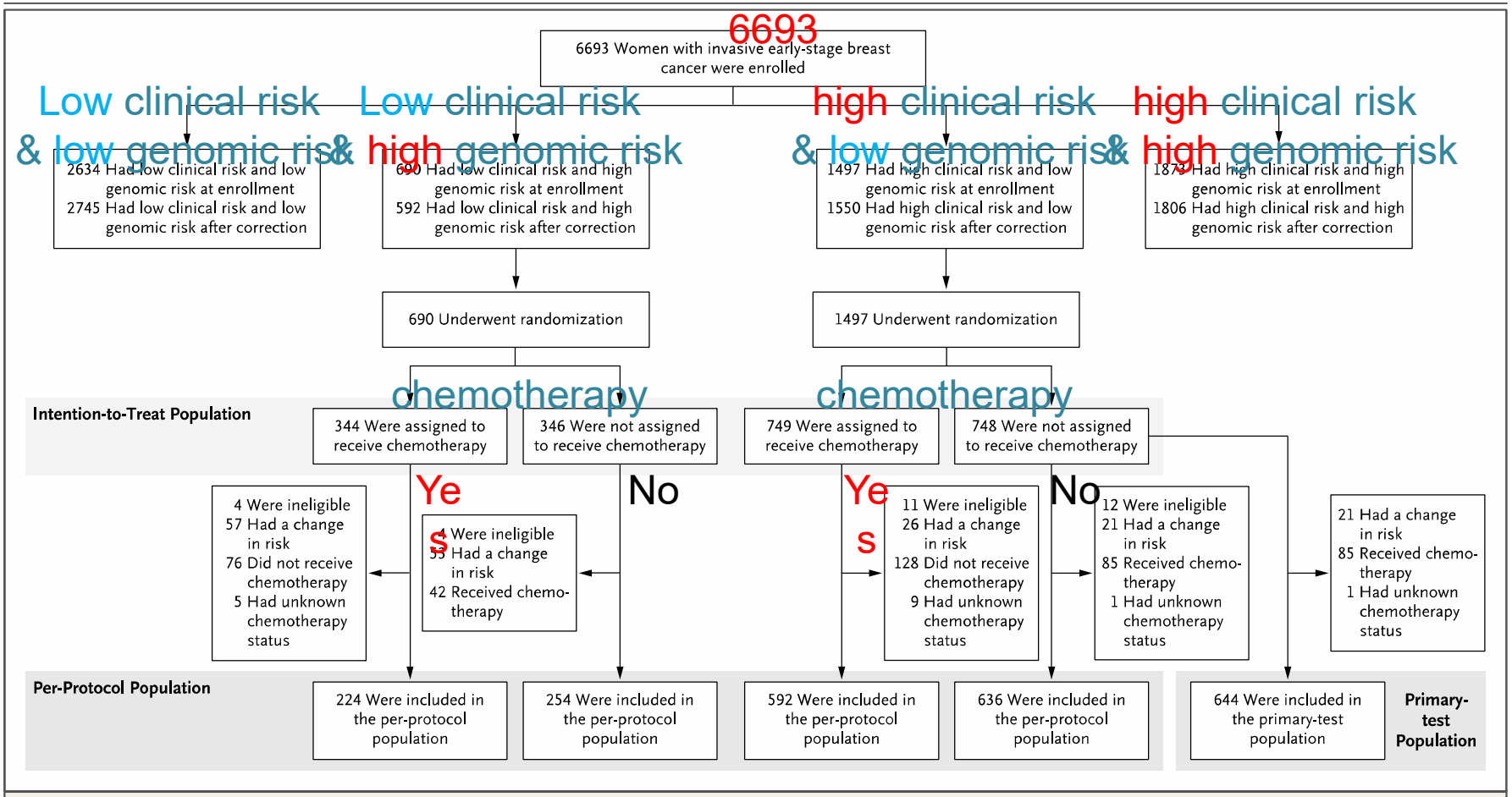
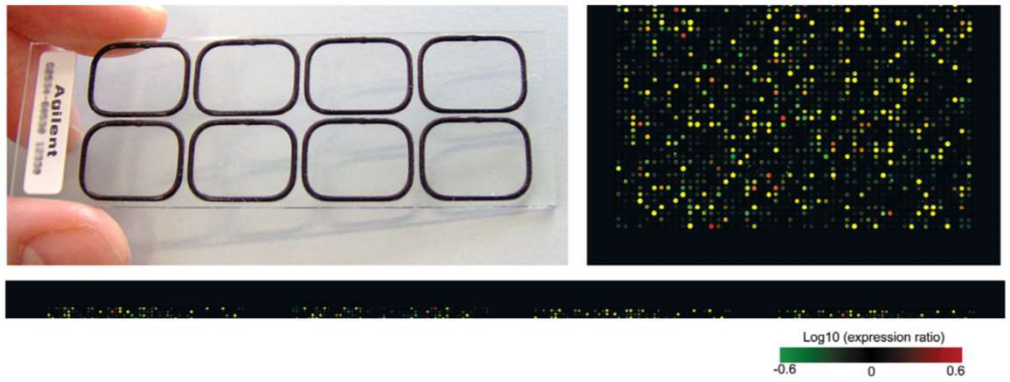
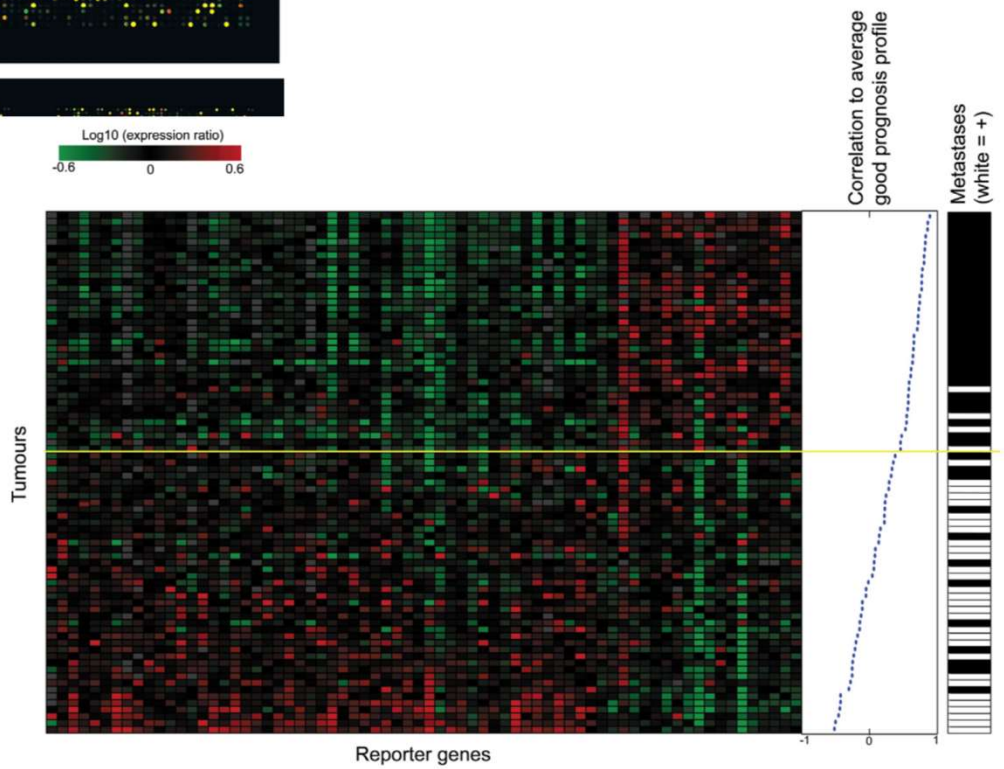
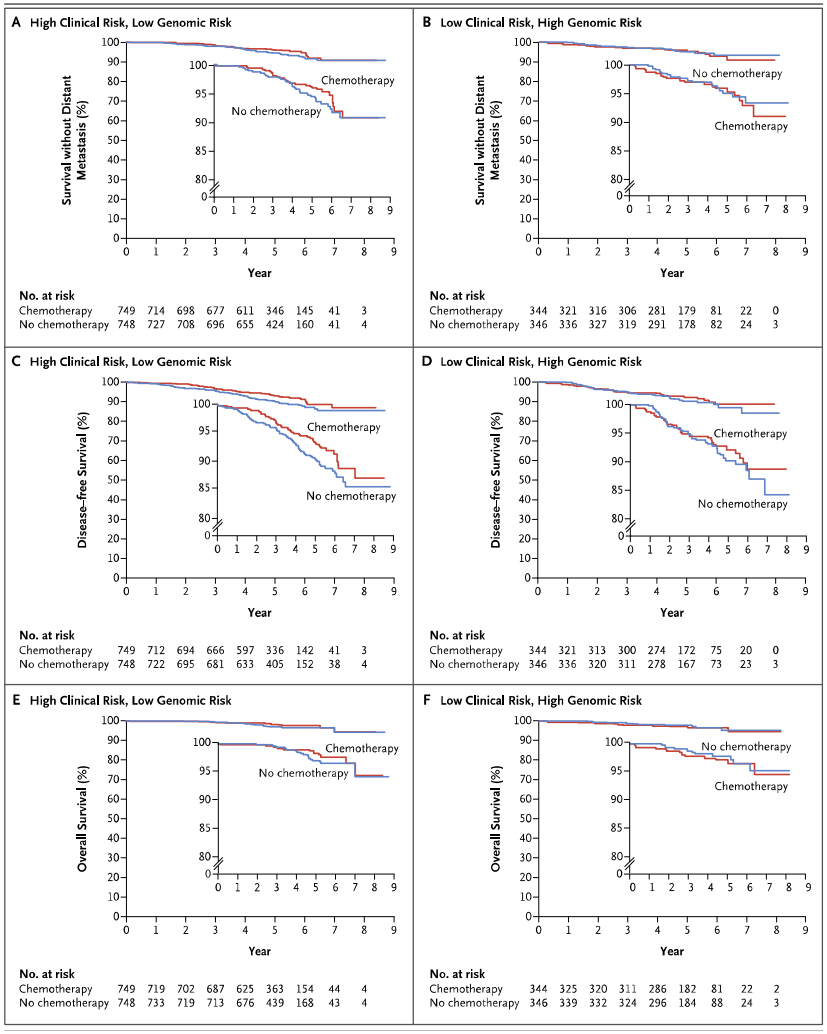


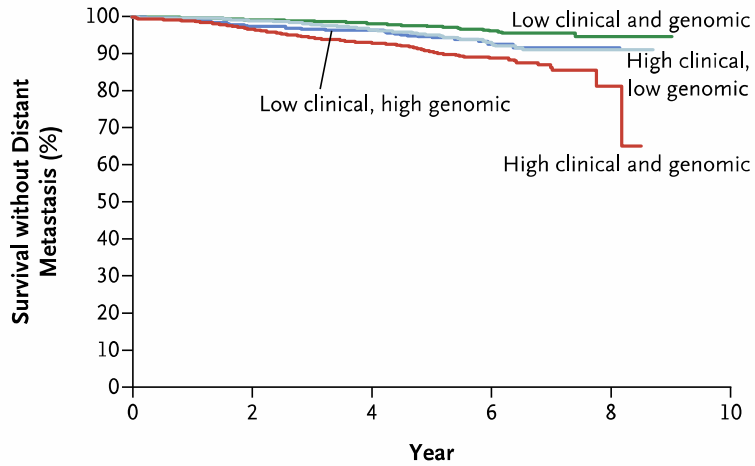
Figure 1. Enrollment and Risk Groups Included in the Analyses.

MammaPrint Test









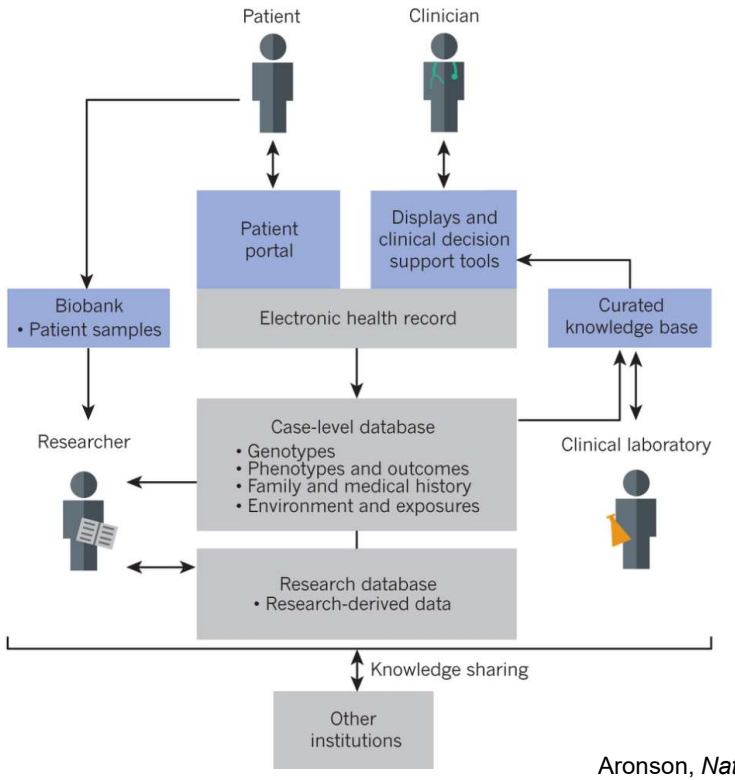
No. at Risk

Low clinical and genomic risk	2745	2628	2331	735	33
Low clinical, high genomic risk	592	550	484	136	2
High clinical, low genomic risk	1550	1457	1317	311	9
High clinical and genomic risk	1806	1689	1462	395	11

Figure 3. Survival without Distant Metastasis in the Four Risk Groups.

The analysis includes all enrolled patients, and the risk groups are based on corrected risk. The time-to-event curves were estimated by means of the Kaplan–Meier method.

Bioinformatics in Precision Medicine



Aronson, *Nature* 2015